CO2: AT-1: Analytical Problem Solving

NAME: K.MOSIKIRAN

REG NO: 192511353

COURSE NAME: DATA STRUCTURES

COURSE CODE: CSA0311

SLOT: B

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Step 3: pop() Removed = 55

TOP

33
66
88

Step 4: push(90)

TOP

90
33
66
88

Step 5: push(36)

TOP

36
90
33
66
88

stack is full

Pointer changes

10. next = 25

25. prev = 10

Delete node = 20

Analysis

Insertion at a specific position : O(n)

Deletion at a specific position : O(n)

Space Complexity : O(1)

2) Perform the following operations using stack. Assume the size of the stack is 5 and have a value of 22, 55, 33, 66, 88 in the Stack from 0 position to size - 1. Now perform the following operations 1) Invert the elements in the Stack 2) pop() 3) pop(). 4) push (90) 5) push (36), 6) Push (11) 7) push (88) 8) pop() 9) pop(). Draw the diagram of the Stack and illustrate the above operations and identify where the top is?

Initial stack (size = 5)

Given:

Index:	0	1	2	3	4
Value:	22	55	33	66	88

1) Pictorially insert an element at 3rd position in a doubly linked list having 6 elements in the list and delete the element at 2nd position from the list. Implement and analyse it.

Initial List

NULL <- [10] <-> [20] <-> [30] <-> [40] <-> [50] <-> [60] -> NULL
1 2 3 4 5 6

(a) Insert 25 at 3rd position.

New node = 25

NULL <- [10] <-> [20] <-> [25] <-> [30] <-> [40] <-> [50] <-> [60] -> NULL
1 2 3 4 5 6 7

Pointer changes

20.next = 25
25.prev = 20
25.next = 30
30.prev = 25

(b) Delete 2nd position

Delete node 20

NULL <- [10] <-> [25] <-> [30] <-> [40] <-> [50] <-> [60] -> NULL
1 2 3 4 5 6

Final Stack

TOP

33
66
88

Final top element = 33

Time Complexity:

- Push = O(1)
- Pop = O(1)
- Invert stack = O(n)
- Space Complexity = O(n) (if an auxiliary stack is used for inversion)

Top = 88

TOP
88
66
33
55
22

Step : 1 : Invert Stack.

TOP
22
55
33
66
88

top = 22

Step 2: Pop()

TOP
55
33
66
88

Step 6: Push(11)

Overflow (cannot insert because stack is full.)

Stack remains unchanged.

Step 7: push(88)

Overflow (Stack is full)

Stack remains unchanged.

Step 8: pop()

Removed = 36

TOP
90
33
66
88

Step 9: pop()

Removed = 90

TOP
33
66
88